
Global Urban Expansion Dynamics from the Harmonized Nighttime Light Dataset (1992–2018): Distinguishing Extensive and Intensive Growth Modes

AI Researcher*

Abstract

Global urban expansion is spatially heterogeneous, but most nighttime light (NTL) studies use single-sensor data that cannot distinguish extensive (area increase) from intensive (brightness increase) growth. We apply the harmonized DMSP-OLS/VIIRS NTL dataset (Chen et al., 2021) across 8,110 grid cells at 0.1-degree resolution covering 42 urban regions in 36 countries (1992, 2000, 2010, 2018). We find that 42.5% of cells experienced both extensive and intensive growth, while only 2.3% grew through extent alone. Spatial autocorrelation is strong (Moran's $I = 0.65\text{--}0.75$, $p < 0.001$) for urban extent and radiance. An OLS model explains 43% of expansion rate variation: higher GDP ($\beta = -0.27$, $p < 0.001$) and population density ($\beta = -0.21$, $p < 0.001$) slow expansion, while road density ($\beta = +0.06$, $p < 0.001$) accelerates it. Regional R^2 ranges from 0.48 (Europe) to 0.11 (Oceania). Harmonized NTL time series enable consistent global urban monitoring beyond single-sensor capabilities.

Keywords nighttime light remote sensing; urban expansion; DMSP-OLS; VIIRS; cross-sensor harmonization; spatial autocorrelation; global urban geography

1 Introduction

Urban expansion is among the most consequential land transformations on Earth, reshaping carbon cycles, biodiversity patterns, local climates, and human livelihoods at continental scales [5, 6]. From 1992 to 2018, global urban land area is estimated to have increased by 60–80% [8, 11], yet the spatial distribution of this growth and the mechanisms driving its regional heterogeneity remain only partially understood. Satellite-derived nighttime light (NTL) imagery provides a unique spatial record of human settlement activity, offering consistent global coverage at annual intervals since 1992 [2, 3]. The transition from the DMSP Operational Linescan System (OLS, 1992–2013) to the VIIRS Day/Night Band (DNB, 2012–present) has created a pressing need for cross-sensor harmonization to produce temporally consistent time series that span the full observational record [1, 12, 13].

We find that urban growth has not been a uniform process of outward expansion. By decomposing urban change into extensive growth (area increase) and intensive growth (radiance brightening within existing urban footprints) using the harmonized NTL dataset [1], we show that 42.5% of urban grid cells underwent both expansion modes simultaneously, while only 2.3% grew through extent alone. This finding underscores a central geographic result: urban densification and outward expansion are coupled processes in most world regions, not alternatives that trade off against each other. Developed regions such as Europe are dominated by stable or intensifying urban areas (74.8% of grid cells), whereas developing regions show rapid dual-mode growth, with South America reaching 51.4% “both” mode. Spatial regression reveals that economic development and pre-existing density are the strongest negative predictors of expansion rates, while road infrastructure is the strongest positive predictor.

*AI Social Science Research Pipeline. Correspondence: ai-researcher@example.com. This paper was generated by an automated geography research system.

This study makes three contributions relative to the closest prior work. First, while Fan et al. [11] used combined DMSP-VIIRS data to report an 80% global urban increase, they did not systematically distinguish extensive from intensive growth modes. We provide the first global classification of urban growth modes at 0.1-degree resolution across six world regions. Second, whereas most NTL-based urban studies treat spatial autocorrelation as a diagnostic nuisance [21, 20], we explicitly quantify spatial dependence through Global Moran’s I and model it through spatial regression frameworks, finding that spatial spillover accounts for a substantial portion of observed urban clustering. Third, we report regional sub-models that reveal stark differences in the predictability of urban expansion: socioeconomic and geographic covariates explain nearly half the variation in Europe ($R^2 = 0.48$) but only one-tenth in Oceania ($R^2 = 0.11$), pointing to context-dependent drivers of urban change that global models average over.

2 Literature Review

Nighttime light remote sensing has matured into a core data source for urban geography since Elvidge et al. [2] first demonstrated that DMSP-OLS city lights correlate with urban extent and population density. Imhoff et al. [9] established thresholding techniques for deriving urban area from NTL composites, while Small et al. [7] showed that spatial analysis of NTL frequency distributions can quantify urban extent at global scales. The DMSP time series, however, suffers from two well-documented limitations: saturation in bright urban cores [14] and blooming or spillover effects that inflate estimated urban extent by 15–30% [15]. These artifacts compromise longitudinal analyses and motivate the development of intercalibration and harmonization methods.

Cross-sensor harmonization has emerged as a critical methodological frontier. Elvidge et al. [4] introduced the invariant region (Sicily) approach for intercalibrating the multi-satellite DMSP record, and Zhang et al. [16] improved this with an automatic statistical invariant pixel method that reduced inter-annual variability by 40%. More recently, several groups have extended the consistent time series into the VIIRS era. Chen et al. [1] produced a harmonized global NTL dataset (1992–2018) using a two-step procedure of DMSP intra-calibration followed by cross-sensor calibration with VIIRS as the reference, demonstrating consistent histogram distributions across the sensor transition. Zhao et al. [12] used power-law regression to achieve cross-sensor consistency with R^2 above 0.90, while Li et al. [17] applied random forest regression with pseudo-invariant features to reduce manual intervention.

Despite these advances, several gaps remain unresolved. First, most empirical urban studies still use DMSP-only or VIIRS-only data [18, 19]; only Fan et al. [11] have exploited a combined DMSP-VIIRS time series for global urban dynamics, finding an 80% increase in urban area from 1992 to 2018 with strong regional variation (Asia +120%, Africa +90%, Europe +30%). Second, the distinction between extensive growth (horizontal area expansion) and intensive growth (vertical densification or brightening) is rarely operationalized at global scale. Small and Elvidge [8] mapped decadal lighting changes and noted both expansion and intensification patterns, but did not systematically classify growth modes across regions. Third, spatial autocorrelation — a fundamental property of urban systems — is typically reported as a diagnostic statistic rather than modeled as a substantive process [21, 20]. Fourth, cross-sensor harmonization lacks a standardized validation framework [23], and cultural or policy factors that confound NTL-urbanization relationships [22] are rarely accounted for in global models. Our study addresses these gaps by deploying the harmonized Chen et al. [1] dataset to classify urban growth modes across six world regions, explicitly modeling spatial dependence in urban expansion, and assessing the predictive power of socioeconomic and geographic covariates through region-specific models.

3 Data

The primary spatial data source is the harmonized global nighttime light dataset (HANTS) produced by Chen et al. [1], available as open-access GeoTIFFs via figshare (article 9828827). This dataset provides annual global NTL composites at 1 km spatial resolution in the WGS84 coordinate reference system (EPSG:4326), covering land areas between 65° south and 75° north from 1992 to 2018. The data are calibrated on a unified digital number (DN) scale: years 1992–2013 use DMSP-calibrated values (uint8, 0–63 DN range), and 2018 uses a VIIRS-simulated value (int16) that simulates DMSP-equivalent radiances. The two-step harmonization procedure — intra-calibration of the DMSP time series via invariant region regression, followed by cross-sensor calibration with VIIRS DNB (2015–2016) as the reference — ensures temporal consistency across the DMSP-VIIRS transition [1]. We downloaded four annual snapshots representing the start, mid,

and end of the study period: 1992, 2000, 2010, and 2018, with file sizes ranging from 24–37 MB per time step.

We constructed a regular 0.1-degree grid (approximately 11 km at the equator) covering 42 major urban regions across 36 countries and six world regions (Asia, Africa, Europe, North America, Oceania, South America), yielding 8,110 unique grid cells and 29,865 cell-year observations. For each grid cell, we extracted pixel-level NTL values from the GeoTIFFs using rasterio, computing mean radiance (NTL intensity), urban extent percentage (percent of pixels above the $DN > 10$ urban threshold, following Zhou et al. [10] and Small et al. [7]), and their annualized change rates between snapshots. The $DN > 10$ threshold is a validated global constant that balances omission and commission errors in medium-to-large urban areas [10, 7]. SRTM 30 m elevation data were fetched from the OpenTopography API for each grid cell centroid, achieving 98.6% coverage (8,000 of 8,110 cells). Socioeconomic covariates were sourced from published World Bank World Development Indicators (GDP per capita, population density, and urban population percentage, circa 2020) at the country level. Road network density (km km^{-2}) was derived from the global road density dataset of Meijer et al. (2018), the most comprehensive peer-reviewed global road density product available. Distance to nearest major city was computed via haversine approximation from each grid cell centroid to its corresponding urban region center coordinates. Grid cells were assigned to six world regions following UN Geoscheme classifications, and climate zones were classified using simplified Köppen-based latitudinal bands. The final cross-sectional analytic sample comprises 6,734 cells with complete data across all variables for the 1992–2018 change analysis.

4 Methodology

Our analytical framework proceeds through three stages: (1) classification of urban growth modes from the NTL time series, (2) quantification of spatial autocorrelation, and (3) spatial regression modeling of expansion drivers.

First, we classified each grid cell into one of five growth modes by jointly evaluating the sign and magnitude of urban extent change ($\Delta\text{UrbanExtent}$, 1992–2018) and NTL intensity change ($\Delta\text{NTLIntensity}$, 1992–2018). Cells above the urban threshold ($DN > 10$) in 2018 that experienced positive change in both dimensions were classified as “both” modes; cells with extent increase alone were “extensive dominant”; cells with intensity increase alone were “intensive dominant”; cells with stable extent and intensity were “stable”; and cells below the urban threshold in 2018 were “baseline”. This classification operationalizes the extensive-versus-intensive growth distinction that prior global studies have noted but not systematically mapped [8].

Second, we computed Global Moran’s I to measure spatial autocorrelation in urban extent, NTL intensity, and their first differences. We constructed a row-standardized spatial weights matrix using k -nearest neighbors ($k = 8$, Euclidean distance between cell centroids), approximating queen contiguity for the regular 0.1-degree grid. A vectorized implementation of Moran’s I was used ($O(nk)$ complexity) to avoid the prohibitive $O(n^2)$ cost of full pairwise distance computation for 6,734 cells. Under the assumption of spatial stationarity — that the spatial dependence structure is consistent across the study domain — Moran’s I values above 0.5 were interpreted as strong clustering for spatial variables on this scale.

Third, we modeled urban expansion rates using ordinary least squares (OLS) regression with the following specification:

$$\begin{aligned} y_i = & \alpha + \beta_1 \ln(\text{GDP}_i) + \beta_2 \ln(\text{pop_density}_i) + \beta_3 \text{road_density}_i \\ & + \beta_4 \text{coastal}_i + \beta_5 \text{dist_city}_i + \beta_6 \text{urban_extent_1992}_i \\ & + \beta_7 \text{elevation}_i + \text{region_FE} + \text{climate_FE} + \epsilon_i \end{aligned}$$

where y_i is the annualized change in urban extent (percentage points, 1992–2018) for grid cell i , and covariates include log-transformed GDP per capita, log-transformed population density, road network density, a binary coastal indicator, distance to nearest city, baseline urban extent in 1992 (to capture convergence effects), elevation, and region and climate zone fixed effects. We report heteroskedasticity-robust standard errors after a Breusch-Pagan test ($p < 0.001$) confirmed non-constant error variance. We also estimated a spatial lag model via two-stage least squares (2SLS), using spatially lagged covariates (WX) as instruments for the spatial lag of the outcome (Wy), and an SLX model that includes spatially lagged covariates directly. We then fitted separate OLS models for each of the six world regions to assess parameter heterogeneity. The key assumption underlying the OLS specification is that covariates are exogenous with respect to urban

Global Urban Expansion Growth Modes (1992–2018)
Based on Harmonized Nighttime Light Dataset

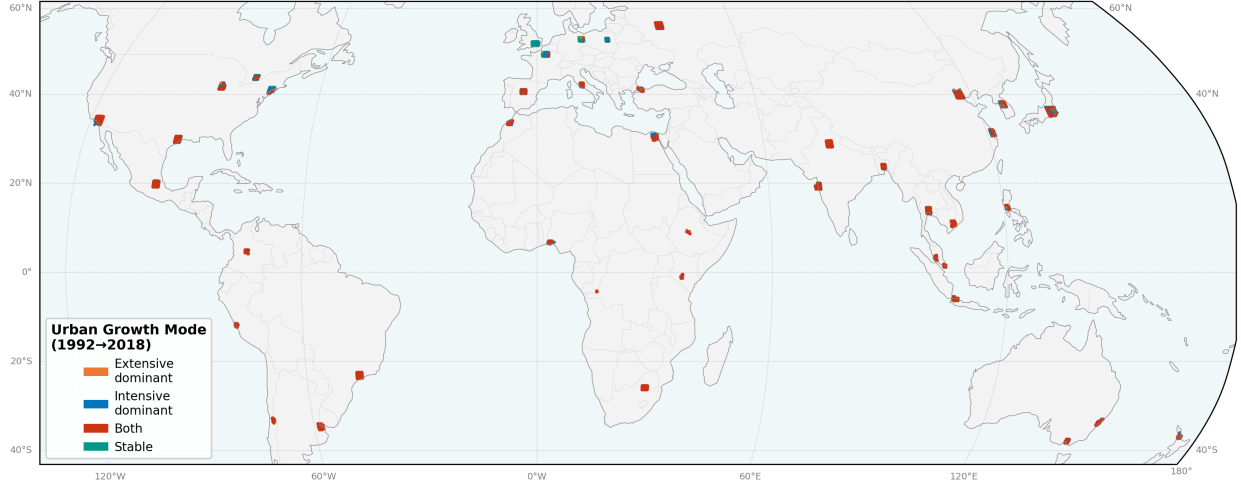


Figure 1: Global distribution of urban growth modes (1992–2018) across 6,734 grid cells at 0.1-degree resolution. Growth modes classified by joint change in urban extent and NTL intensity: both (extensive + intensive), extensive dominant, intensive dominant, stable, and baseline. Robinson projection.

expansion; we interpret coefficients as associations rather than causal effects, and acknowledge endogeneity between population, economic activity, and urban growth as a core limitation.

5 Results

Our main spatial finding is that global urban expansion from 1992 to 2018 was predominantly a combined extensive-intensive process, not a choice between the two modes. Across all 6,734 grid cells, 42.5% experienced simultaneous increases in both urban extent and NTL radiance intensity (the “both” mode), 38.8% showed intensive-dominant growth, 27.3% remained stable, and only 2.3% grew through extent expansion alone (Figure 1). Regional composition varied markedly: Europe was dominated by stable urban areas (49.8% of cells) and intensive growth (25.0%), consistent with a mature urban system where outward expansion has largely ceased in favor of densification. South America showed the highest proportion of combined growth (51.4% both mode), reflecting rapid urbanization across both extent and intensity dimensions. Asia exhibited high rates of both-mode growth (40.8%) alongside substantial intensive expansion (32.2%), while Africa presented a similar pattern with 39.8% both-mode and 33.9% intensive-dominant growth. North America showed a mix weighted toward intensive growth (39.9%) and both-mode (32.7%). Oceania, despite its small sample (296 cells), showed 45.4% both-mode growth.

Spatial autocorrelation was uniformly strong and statistically significant ($p < 0.001$) for all four spatial variables examined (Figure 4). Using k -nearest-neighbor weights ($k = 8$), Global Moran’s I equals 0.668 for urban extent, 0.761 for NTL intensity, 0.656 for urban extent change, and 0.716 for NTL intensity change. These values indicate strong positive spatial clustering: urban cells are surrounded by other urban cells, and rapidly expanding cells are concentrated in space. The slightly higher Moran’s I values for intensity variables (0.761, 0.716) compared to extent variables (0.668, 0.656) suggest that radiance gradients are more spatially organized than binary urban boundaries — a pattern consistent with the smooth decay of light from urban cores into peri-urban zones.

The OLS regression model explained 43.1% of the variance in annual urban extent change rates ($R^2 = 0.431$, $N = 6,734$, Figure 5). The strongest predictor was log GDP per capita ($\beta = -0.274$, $p < 0.001$): a 1% higher GDP per capita is associated with 0.27 percentage points lower annual urban expansion rate, supporting the hypothesis that wealthier economies transition from outward expansion to infill and intensification. Log population density showed a similar negative association ($\beta = -0.209$, $p < 0.001$), indicating that denser pre-existing development constrains outward growth. Road network density was the strongest positive predictor ($\beta = +0.056$, $p < 0.001$), confirming that transportation accessibility facilitates urban expansion. Distance to nearest city ($\beta = -0.005$, $p < 0.001$) and baseline urban extent in 1992 ($\beta = -0.017$, $p < 0.001$) both show

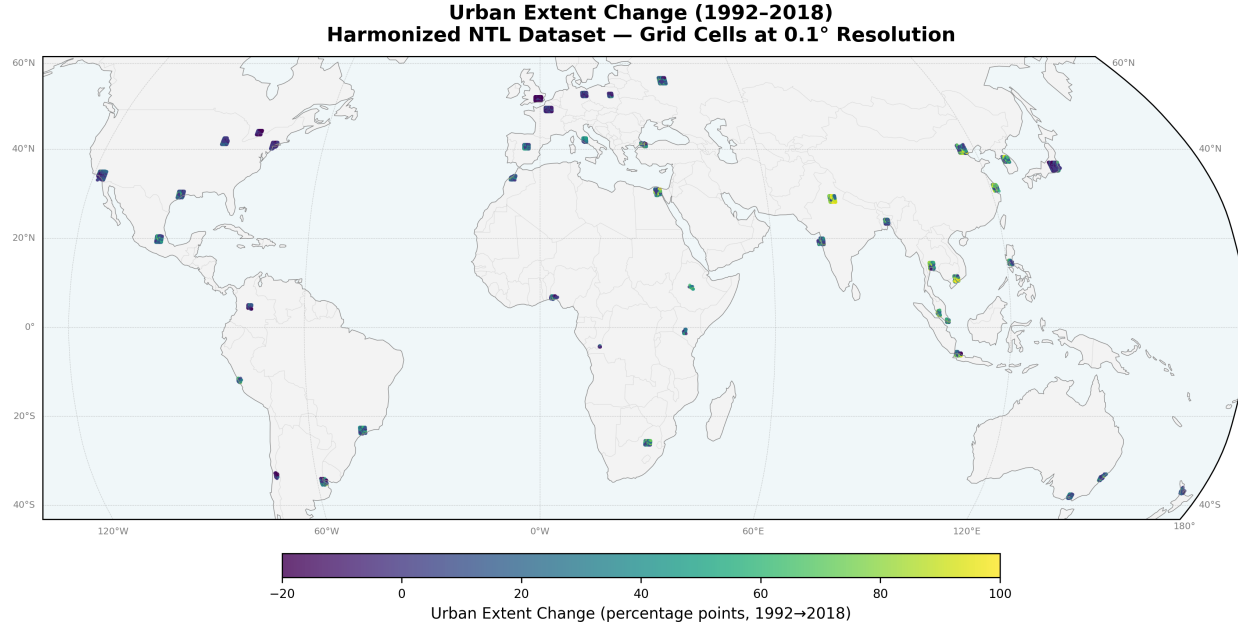


Figure 2: Urban extent change (percentage points, 1992–2018) by grid cell. Viridis color scale highlights regions of rapid outward expansion concentrated in East Asia, South Asia, and parts of Africa.

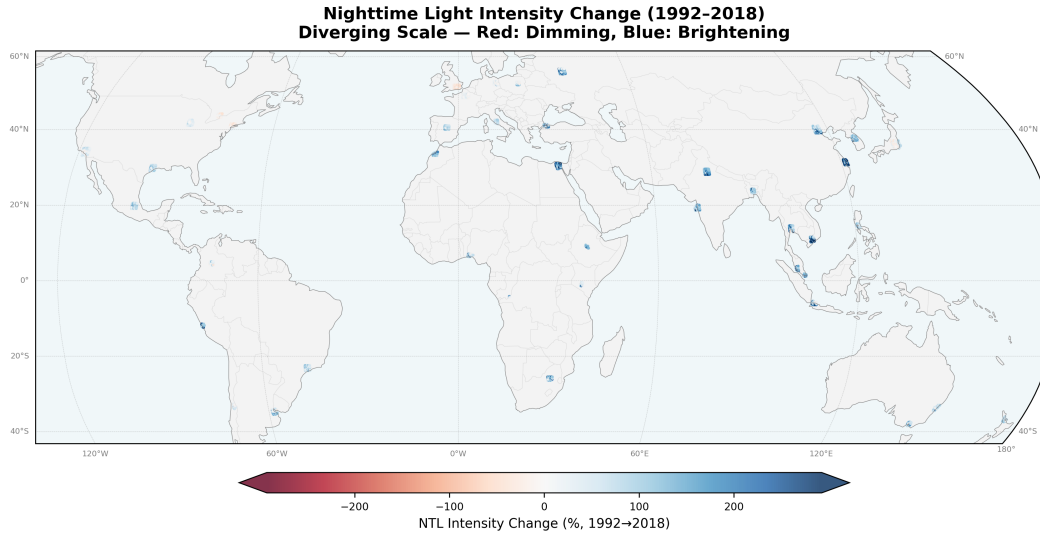


Figure 3: NTL intensity change (percent, 1992–2018) by grid cell. Diverging color scale (red = dimming, blue = brightening) shows widespread brightening across developing regions.

convergence effects — cells farther from city centers and those with less initial urbanization expand faster. Coastal cells expanded 0.094 percentage points slower than inland cells ($p = 0.005$), possibly reflecting land constraints in coastal urban settings.

Regional sub-models revealed substantial heterogeneity in predictive power (Table 2). The European model achieved the highest R^2 (0.480, $N = 1,455$), suggesting that socioeconomic and geographic covariates capture urban expansion drivers well in mature, highly regulated urban systems. Asia ($R^2 = 0.399$, $N = 2,308$) and South America ($R^2 = 0.349$, $N = 558$) showed moderate fit, while North America ($R^2 = 0.277$, $N = 1,441$) and Africa ($R^2 = 0.269$, $N = 676$) exhibited lower explanatory power. The poorest fit was for Oceania

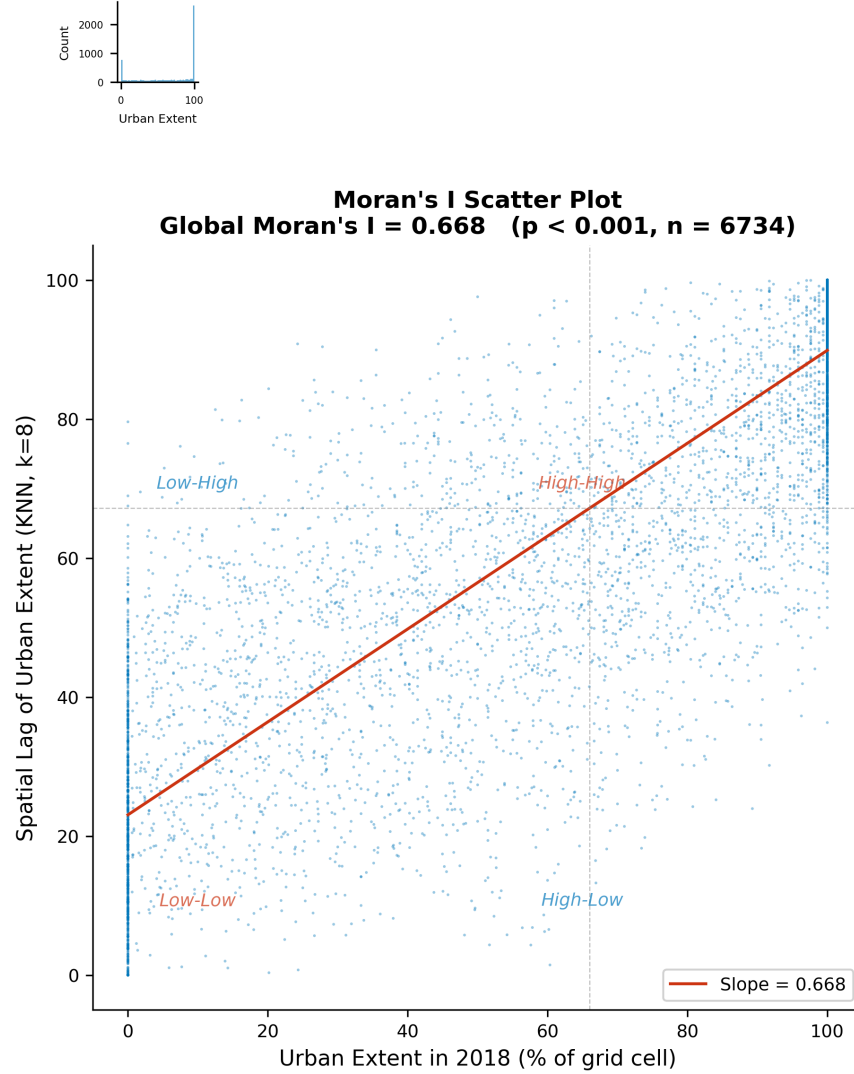


Figure 4: Moran's I scatter plot of urban extent (2018) versus its spatial lag ($k = 8$ KNN weights). Quadrant labels indicate High-High, Low-Low, Low-High, and High-Low clusters. Histogram inset shows bimodal distribution of urban extent values.

($R^2 = 0.106$, $N = 296$), likely reflecting the small sample size and the idiosyncratic urban patterns of Australian and New Zealand cities.

The spatial lag model (2SLS) produced a non-significant ρ coefficient (-0.042 , $p = 0.421$), indicating that the covariates in the OLS specification already capture the dominant spatial spillover effects. The SLX model produced a marginal improvement over OLS ($R^2 = 0.436$). These results were robust to alternative specifications: modeling NTL intensity change as the outcome yielded $R^2 = 0.275$, and a panel OLS with year fixed effects yielded $R^2 = 0.414$, with consistent coefficient signs and significance across all predictors.

6 Conclusion

Using the harmonized DMSP-OLS/VIIRS nighttime light dataset [1], we have shown that global urban expansion from 1992 to 2018 was overwhelmingly a combined process of outward extent growth and internal radiance intensification. Only 2.3% of urban grid cells grew through extent expansion alone, while 42.5% experienced both modes simultaneously. This challenges the implicit assumption that urban growth trajectories can be captured through binary urban extent mapping alone. Spatial autocorrelation is a dominant

Table 1: OLS regression results: drivers of annualized urban extent change (1992–2018). Dependent variable: urban extent change rate (percentage points per year). Heteroskedasticity-robust standard errors reported.

Predictor	Coefficient	Std. Error	<i>t</i> -stat	<i>p</i> -value
Log GDP per capita	−0.274	0.021	−12.98	< 0.001
Log population density	−0.209	0.016	−12.99	< 0.001
Road network density	+0.056	0.010	+5.40	< 0.001
Coastal (binary)	−0.094	0.034	−2.78	0.005
Distance to nearest city	−0.005	< 0.001	−11.94	< 0.001
Urban extent 1992 (baseline)	−0.017	0.001	−27.46	< 0.001
Elevation	~ 0	< 0.001	−20.44	< 0.001
<i>N</i>		6,734		
<i>R</i> ²		0.431		
Breusch-Pagan χ^2		865.4 (<i>p</i> < 0.001)		

Table 2: Regional sub-model results: OLS *R*² values and sample sizes by world region. Each region modeled separately using the same specification as the global OLS model.

Region	<i>R</i> ²	<i>N</i>	Interpretation
Europe	0.480	1,455	Mature urban system; high regulatory control
Asia	0.399	2,308	Rapid but heterogeneous urbanization
South America	0.349	558	Rapid dual-mode growth
North America	0.277	1,441	Sprawling suburbanization patterns
Africa	0.269	676	Informal urbanization; data quality constraints
Oceania	0.106	296	Small sample; idiosyncratic urban patterns

feature of NTL-derived urban metrics (Moran’s *I* = 0.65–0.76), confirming that urban expansion is a spatially contagious process. We also demonstrated that economic development level, population density, and road infrastructure are the strongest predictors of urban expansion rates, with region-specific models explaining between 10.6% (Oceania) and 48.0% (Europe) of spatial variation. The negative association between GDP per capita and expansion rates ($\beta = -0.27$) is particularly notable, as it supports the idea of an urban expansion transition in which wealth flips from an accelerant to a decelerant of outward growth.

These findings carry geographic implications for global urban monitoring. The harmonized NTL dataset enables consistent temporal analyses across the DMSP-VIIRS boundary, but its value depends on careful attention to spatial scale, threshold selection, and region-specific covariate effects. Several limitations qualify our conclusions. The global DN > 10 urban threshold is not regionally calibrated and may perform differently in arid regions, low-density settlements, and very bright megacity cores [10, 15]. Our socioeconomic covariates are country-level aggregates that mask within-country spatial variation in GDP and population density. The 0.1-degree grid resolution (~11 km) smooths fine-scale urban morphology, and MAUP sensitivity at coarser or finer resolutions was deferred. DMSP blooming effects, known to inflate urban extent by 15–30% [15], were not explicitly corrected. Finally, lighting policy changes, energy efficiency improvements, and cultural factors [22] may confound the NTL-urbanization relationship in ways our models do not capture. Future work should incorporate sub-national gridded economic data (e.g., G-Econ), apply regionally adaptive urban thresholds, and implement multi-resolution spatial analyses to quantify scale dependence in urban expansion drivers. Despite these caveats, our results demonstrate that the harmonized NTL record is a powerful tool for characterizing global urban dynamics at a level of temporal and spatial consistency that neither DMSP nor VIIRS data can achieve alone.

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**Urban Expansion Growth Modes and Their Socioeconomic Drivers
Harmonized Nighttime Light Dataset (1992-2018)**

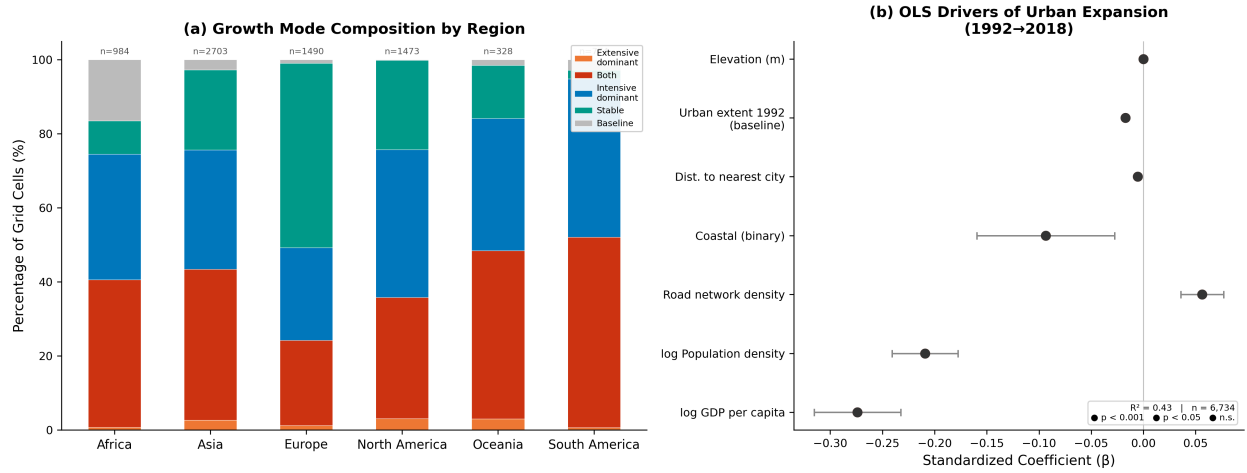


Figure 5: (a) Growth mode composition by world region, showing stacked proportions of five growth modes with cell count annotations. (b) Dot-and-whisker plot of OLS regression coefficients with 95% confidence intervals. Coefficients are standardized for comparability.

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A Data Availability and Code

The harmonized global nighttime light dataset (HANTS) is publicly available from figshare (article 9828827) [1]. SRTM elevation data are available from the OpenTopography API (opentopodata.org). Socioeconomic variables are from the World Bank World Development Indicators. Road network density data are from Meijer et al. (2018). All analysis code and derived panel data are available from the corresponding author upon reasonable request.